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P2P Settlement — Wallet-Native Cross-Rail Swap

What this is. A peer-to-peer swap between crypto on **Bigtangle L0** and fiat on **PayPal** (or the CNY rails — WeChat Pay / Alipay / bank). The seller escrows tokens in a **2-of-3 P2SH address** (seller, buyer, engine); the buyer pays an exact-amount fiat obligation; the engine proves the payment and the escrow releases. Every irreversible step is signed with the wallet's own PQ key, and the engine can only ever **co-sign** a spend — it never holds your funds.

Where. The **P2P** screen in the wallet (sidebar → Trade → P2P). Everything below is captured from that screen on the local demo build.

Demo vs. live. The screenshots come from the demo build: the engine runs in mock-PayPal mode (stubbed invoice URLs and synthetic transaction hashes) and the lock/verify/release legs are signed with a local engine key. The state machine, the signatures, the append-only event log and the on-chain audit anchor are the real thing; only the external PayPal calls and the L0 broadcast are stubbed.

No manual "Confirm". There is no seller "confirm" button on the PayPal rail — the engine verifies the payment from its own evidence. (The CNY rails, which have no webhook, use an explicit seller confirmation instead; see `docs/p2pcny.md`.)

The flow at a glance

ACTIVE —match→ MATCHED —lock→ ESCROW_LOCKED —payment→ PAYMENT_PENDING

|

PAYMENT_PENDING —verify→ PAYMENT_VERIFIED —release→ ESCROW_RELEASED —payout→ COMPLETED

#	Step	Who acts	On-chain / engine
1	Seller lists a signed sell order	seller	order stored, not funded
2	Buyer matches (receive address + PayPal account)	buyer	swapId created, escrow address derived
3	Seller funds the 2-of-3 escrow, engine proves the lock	seller + engine	ESCROW_LOCKED on L0
4	Buyer pays the exact-amount invoice and reports it	buyer	PAYMENT_PENDING
5	Engine verifies the payment itself	engine	PAYMENT_VERIFIED
6	Engine + buyer co-sign the release, tokens move	engine + buyer	ESCROW_RELEASED
7	Engine pays the seller via PayPal Payouts	engine	COMPLETED

Every transition is anchored as a `social.p2p-swap` record on the L1-SOCIAL chain, so the whole lifecycle is publicly auditable without exposing either party's PayPal details.

1. The seller lists a sell order

On the **Open sells** tab, fill in the order: the token and amount you give, the fiat price you want, the currency, the chain the tokens sit on, and the ****payment method**** (PayPal or a CNY rail). Terms are locked into the order when you list it, so this is signed with your wallet key.

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List a sell order

Token

USDT

Amount

10

Price

10.10

Currency

USD

Chain

L0

Payment method

paypal

wechat

alipay

bank

List order

No open sell orders

Once listed, the order is `ACTIVE` in the engine and appears in the public order

book (it carries no personal details):

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My swaps

Refresh

List a sell order

Token

USDT

Amount

0.00

Price

0.00

Currency

USD

Chain

L0

Payment method

paypal

wechat

alipay

bank

List order

10 USDT

10.10 USD

Chain: L0 - paypal

Chain: L0 - paypal
did:key:z446Gmp1FqLFYjzauMn975CaXnnFDgFvZy8da4bnWbaJ6qrmw0Q7nlwkjYtVvKPF6WnRRR7JvjQEw6L2gK1uBzymGNFw85sXKQEr1cwkZp2vq6Xhy5Z8ARZyWakAZufyWbELKjgC3uJgxpEhVYjUzcp1HptLHtU1yU2E3u0KXfZ5QudPmjADayhtVqT1ha5fMeBguGVYvYtU5G6RR99s3v7tVw6LyygAaQWppgQ9PQEGTCdWRUpJXzohJRDw7nDLULR8FZgposy9B6Zv4PHEQZ3SLAEBCuAm07vA8TromhyJPrJgyXXM1Cobjst7a7YXfB99nLEhyvAZ71mlk2k9PUH9eAc6ZkuTDFRSQUJwJGhSCDFwyz6VP2oREfqbzJkdqalukcF5pW83sFIdGJvSdnqPKKh762c2e7oFNN31G1tFygc17IHv55Mnmg652oNTXFFNsUdK22zpfGyTJHNjXEPd651vCpt2uLBNj1z1JhsaTHKwHw2VnJL3vHMAUC2NM525dv3vq8f5Upa7Tnq3525zxrVuZCW3XP5Dh+LLuHKTbJ3d1sT2wufJAAnEazg3D0QKEJ3Q9A8mZ5Nhh4PmAm13TJm06u3BRN2X4ByAJ1rsL0LTCPEuHEkwiKzyyLWbaw5BaxShxu4qz89Wao89Sal9B6GGBFHT1RLVVYnXywg9G9npJCfQMK2NJZe09EdhHJUp75KjY3BqaDya8ERHAJ360EY5FTE85kKHCHBJCUCedolmKNQYCEZBaxcTKVrZ4b0UTCJ5w63tRSDPHfajTL29jn2tYG3VPLDurg5rbaEfF8Z3WrcYUgnmbV08BcgDGeprf61rWm1g7Ru8CwdrXxaAeSohaU9FDV4AvpqtRghA1T4BaFvegdEhVzZwCkncp4L8sRf6GkuErlq6BYCJSLFADTyP18xq8bdrP9GR6oHJ3chJyW612tT4eMQV650tpodlekAYE5uhs5yALptc60TffpArd7JFRUWLGhnyEoHTYkxv8ndrJ6c94Zgso3wK0gH0PKF83fm4xkhZ4zh93x4m5cF6sGN2jwhs9F4rYU0W96k93F2AGosb1ZLsZAJttrhan41uFwrf8HWPJ9t3aZofHghq4eTdhbHhQ82oTWUdetnnaGhwqTwdRhyupux7wzsmkkyXghEsZ9eYm8xFpBlhVekL1fotn8Z280ZF11UFwdsZZ1InqH1TuLcNrL2VfBx1SR2rnVzEcTkwA55CnF8JPray0H0e0JfJmK2ubwBwWfcmqzE9BjgPv5gcpAQ6AXXicZhgIm93IzFM4BwHicGuc5YeCnPdN5E3dmgsVmrXng9BFR21hrXyMR03TNT1251Z1Cpek3AnugtY55xR8J4bzxasvsZL5CiwSgglSRRHpvnlKh2VqvYagz9Pm3p5uz2hHrcAvOnpL8kUwFyeb8r2WFlr5glCNXcbZV3n5WqM1YbLgh5f8eRHBZvzyndp5yXo5Am1JxDNEJpVpKEACmpM6KndkLHWsGEdNCAqCF5RDsw5p6WLiXCkagh731Y5H6CukpKw2ubPocX4yz47J3J3CLJNtMwH7906f1c1VCJ2Yq8D8gULfDYFYTOTYeu49PQCHYUp61P9u45w7K02cTKqLufFoEThtgBA329d9GTPKPs4D8ByTKTGF17xSeel8bJegOC1ftGn43HV83oMuZVym0aJa8H3e1RUPKkhrCzAEiEy8GcxZJ0GdmKJum09VHuhrrh3J8JtXS1WsNtnjKJfFaaQ0GRh1YvrcdyAqUsgqKHdYdLnzrpyrRH4M4F13cAVQHT5KwC7urccNEN3EH92BTHoAbFGX3sewP0u2T1gH86ztPw79YrYvz5P1ZJnD9H2edv1Y1CJRS1fyhgRtWfVxybsAhtcbG4c1xRbB31JecU6m6Zmw1J7cUHN2Rrrrr24e98CcU13tqv8fJ0CnzH95o4FbD7YG537ka8ZMSNg25Jup1x8w0JNUAE3b1ds1u5MpnR6sR1YT2NhC1JDtP1LPufWutAk49jxjBq7mDP0aqBfdWjQpEshNhybdfDsyjCE267sy09TA9DE1EtozwfxYfkTckfK2cAqYghuQHFEGv96XCLYR6Gvm6yFRXUft5NMM1JyJH9zC1b7CRYapoX0EuAvFw13aNECxfT1zD1b5Z5zbZ0AvacPnL9U5mr

2. The buyer matches

The buyer opens the order and supplies the **receive address** for the released

tokens plus the **PayPal account** the invoice should bill (and an email for the

invoice). Matching commits the buyer to pay, so it is signed too:

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← P2P

Chain

L0

Payment method

paypal wechat alipay bank

List order

10 USDT

Chain: L0 - paypal

dId:key:z446GmP1qLpYyZauMh975CaXNhFDgFvZy8da4bnWbAJ6qrmdQ7nHwkjY1VtKPF6WnRRR7JvJj0EW6L2gK1UzYmGNFwW85sKXOErlCwkZp2wq8Xhy5ZBAR2yAMAXZurYwDELKjgc3u2xpndhVYtUzCp1MPLHtU1yU3E3u0dWfZ5QudPmJADAhYtVg1ThsFWeBgUGVYtUS6GR9B93v7kVW6L1yqAaQWpQVQVQEG7CD0wDpXZehfRbW7nDLULrBS7ZosvqB8Zv6REQZ3SLAEBCuAmD7VADTrmHyJP3JpYXW1c0b157w7X7YB99LEHyvAZ71mUk2K9PUH9eAc6ZkuTDFRSQUuWlGhSCDFwoyZ6zVP2oREfqbJkdqalUkcF5pW83FIdGJVadnqPKH7b2c2e7oFN31G1Tfygc171HV55PmMG652dNTXtFFNsUdK2J2pfgYtJHnjxEPDe5lvCptZuLBNj1zJjhsatHkMhWgZVnJL3vHMAUC2W6525dv3WgBf5UpA7tnq3525zxrVwZCW3XP5DhTLLqHk7h3d1T2wofFAAeAZg300KCEJDP9B8m73Nhb4PmW6137W606ao18W0zX4BvAJ1rLdL1tCEPHEkLwZxyvLWbns5BaxShou4qz89Wao89Sal986gBPHt1rLVVYmXywg9G9npjCFQWkZUJZEo9eEdHhJUp75KjY3BqaDyARERhAJ360EY5FTE85kKHChBJCuREdoLmKQNYCEZBaxCKTVrZx4bUTCj5W6zTRSDPMfabJtL29jn2tYG3VPLDUrgM5rbaEfT8ZMwrcYUgmBV088cgDgepr651rWnR1g7RuBdUdRxxAXEsonH95FdV4AvpqfRgAAT14bAFveqB8EzrZwCCKncp4L8sP6sKueErlq6YCljSLfADTQPIlKxq8sDpP9GR6GhR3ch3yW6J2L74mPQV6SQtppdAaKAY5uKsSyALtpc60TffpArD7jFRUWlghnyEohTYKxw8ndr36c94Zgso3wK0gHOPFKB3fmdxkhZ4zh93x4n5CF6sGNZj3whs9F4YU0W69k93J2AGosb1ZSLZAJtRfq41uTvrF8MWPJ9t3aZofHghq4etDhBHQ82oTW0DettnaGhwqTwDrHyupux7wzsmXkyXghEsZ9eYmnbxFPb1hVEX1fotnB2Z8DZ7F1UFWd5ZZ1JnpH1JULCefR1ZvTb1382NzVzCEtKwAs5nFBf3PpYdQDe6tJfUKJubwbnW6KmcqE5BqjtpP5qcpQ6AXXUc2BpIn91ZFM4BvH1cGuc5YecnPdXN5E3dmgsvWrrXng9BF821hRxyMR93TNTF251Z1Cpek3AnugtY55xR83abzxasvsZL5C1w5gglr8RHpvnlkh2VqvYagZ9Ppn3puz5aHkrC5AvQWpBbkWfYeb8r2MF1r5g1CNXcb8ZV3n5WqM1YbLGH5f8oRHRZvryndp5yXo5Am1JxONEJpvpKEACmpM6KndkLHkS6EdNCaQcF5R0Sw5p56W1XKagH731Y5f6CupkR2jubaPbK4yx4K7JHJChLJttrHwH796fLc1LVCjYagD8gLLfDYFXTOTYeu4h9PQChNYP6o1P9y45w7820c2KqglDfPEtTngtBA3Z9d9GTpKp540ByTKTqf17x5eel8bjegOC1fT6n43HV83oMuZvmQa3aB93e1RUPkhrCzAE1Ey8GcxZjDgdmKJum09Whurhrh3JdJfX51W5NtnjKJfFacRQGRh13YrCdyAqUsqgkHYdTLnzrpyrRH4bM4F13cAVQHT5KwC7urccNEN3EH92BTHa6fGx35ewPduZ71gN86zPtW79x7ryvz5Tw1ZnD982ecdv1YL1JLRS1fYhR8fthPvxybsAhtcbG4jsRB8J1Jnc0M6mZmW1JCUNZTRrrrs24e98Ccu13tqvo8FjDcnzH95a4b7G057Ka8ZM5Nq25Jup1xbWQ1NUAE3bJds1u5Mpn9H6sR1YT2Nhc1D1P1LPUFWutAkD9jxJbq7mDPbaqBfDwJQpEShNhybdfD5ysIC6267syo9TAdGE1EozTwfYfktwckFK2cAaqYghuQ0HfEgXv96XCLYR6GvmyfMRXUftc3NWM1JyJH9Zc1b7CRYapox0EuavAkw13aNECxf1LzD1b525zb2o0AvacPm199UsmrLmB7Eden1fgTxxJ8HhgyrkaA7T79H62ys5G6HdcJAFRAKLV5ekH5AVcKssyVK3f3jphT53dmhvr13E3Na1pZ2uP2J1083c1fFwef9Hw1z1c36lwoFmJ39opve1b31K25E2VKCVPPFPX85h1K9sbcp1F2sH5jyczwQC2ty3uvFudn8nbk8p0nNMhvbEQ74H56dKVPzxMba1aeZ98W5N3W3LvZovtyq3E4zxz0QURj2s43j3p3uT63xeAT1gKvqLqgE9nc2pmgRBRREt3jD3hnszvz2cMgZLufRbBPESfzX94Jct6nG6Hj2GmcuG9WbdlwqRm1RNY9ecvQdUfK0RLf5m6j55CVPFFj0pJCeHqpPZ5g9H70ZmncPFR4WRHTDnS82o7dLZzmh18Dh8s1mCakayof1u0eVku130614z5HfYg6s9VXXE6vHfxRJExrFmWkxZuRSqgdGhct5eHUX1X1ctVWNZUEJ5A915HorDTZGCRmPUaNA6u7Yc5fX5NWCf8o7Tnu6SQWU2MhLm0aeQPOLJ6Q58jUv4psZ4LhZG8J7JNFAweaKA3Da4qVnXUp7HUAkX1GG5kK9x2L1Z27G1ggYpMSHYwYPFJZcRkufn58ogTmgnXQXaXoPLDJSRBH8C2m0CuzU7V96KLQ4M4Qe094EZLV11Q25H9g45oxTNG5UyT7e14jnfqWcJNzg15kH8KAvq54uWbFTTEQ521ZHHh8GfPCbnEKZAv6WkU75YgmllJ1HSEfTskstYtT4fIZN24ETTox4cAH6GdHm48ZdmVXFub7H6hZf8kL438WkAvKctyY3VavVj0bVd5E8UAVB9PQK603o3d4854570Dpx1PXxfRbCqdX9h3p9U3eqfag4UcHlorGukhejKkEX4UftaDf8K10aaJjRmUSDcfamDuefCev71LhnpCFRpABz8UvNZ3y6KRWwW6tsh8tN5dGk6GAU2B5DyryXRFgqXDP5vKwq651NF2ve5JwCfRuH5COFz6PEdZ1fPHWeyWtPbQ44Y2T6wa1Lxb19R8U2ZkX832Ydwy9G0B45F7K9LpHk8KwCvXo4baudYK4CQWQVjketrc1p5fag79D1T7n7jg677KwMeaBfR3M6b8up2x256a3MR8s8d6JkruLUGfh3j8cTTT9a4m2zs53AKXggH5Wv9W0U1WpCMu2WHSjW3D17sd3eASmucupoda3paQl4puRPJvtqNUSwCASDqmhMa7Hnec

Receive address

BP2pDemoBuyerReceiveAddress

PayPal account

buyer@example.com

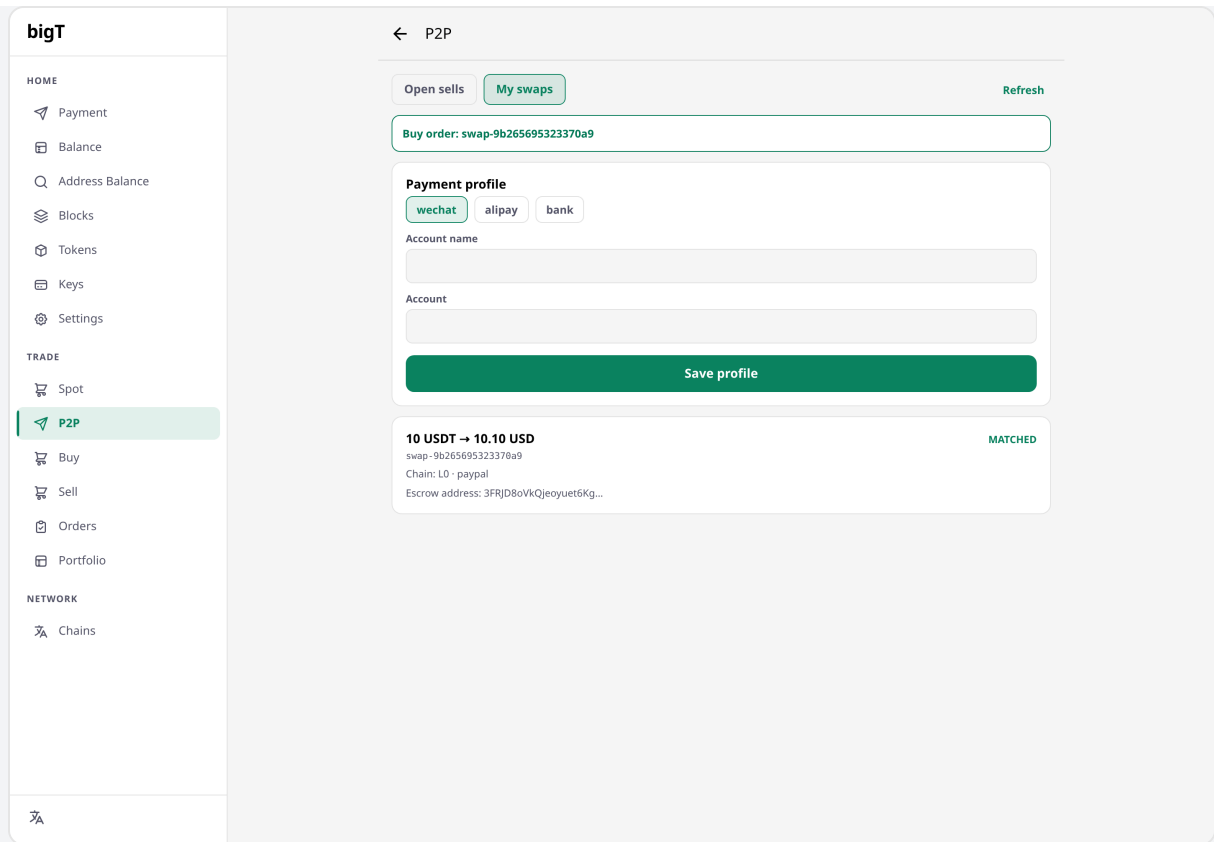
PayPal email

buyer@example.com

10.10 USDT

The engine creates a unique `swapId` for the lifecycle and the swap moves to

MATCHED , shown on the buyer's **My swaps** tab:



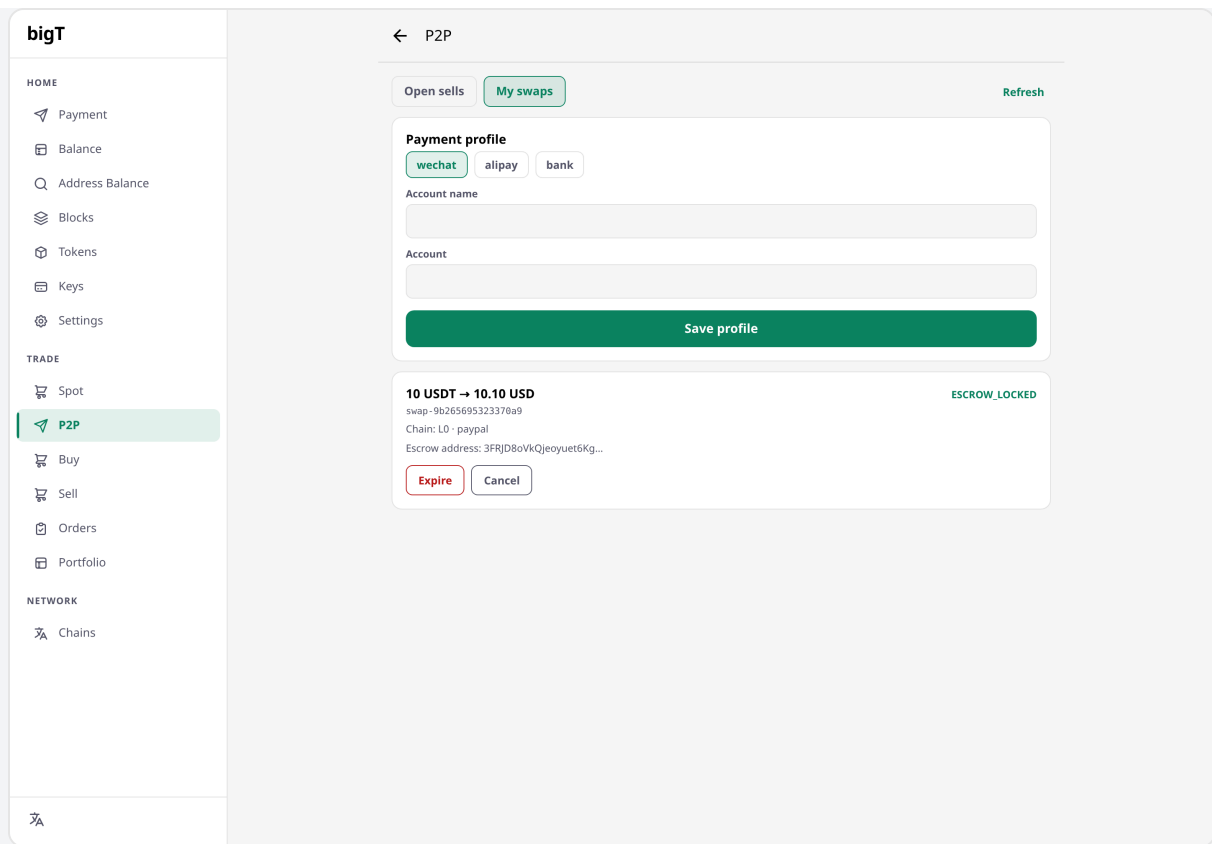
3. Escrow lock – the seller funds L0

The seller sends the tokens to the deterministic 2-of-3 escrow address and

reports the transfer's `txHash`. The engine proves the lock from the chain

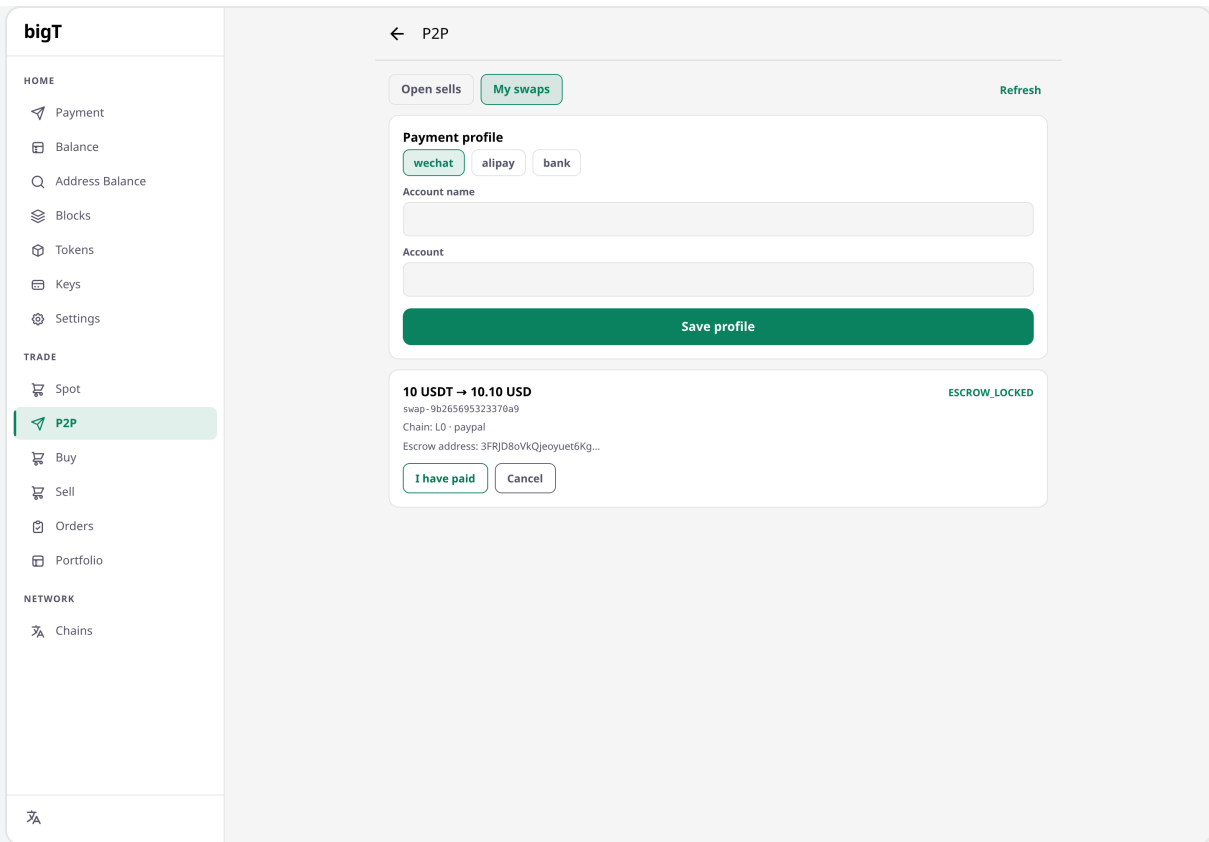
(`CONFIRMED`, destination `escrowAddress`, amount) and fails closed if any check

does not hold. The swap is now `ESCROW_LOCKED`:



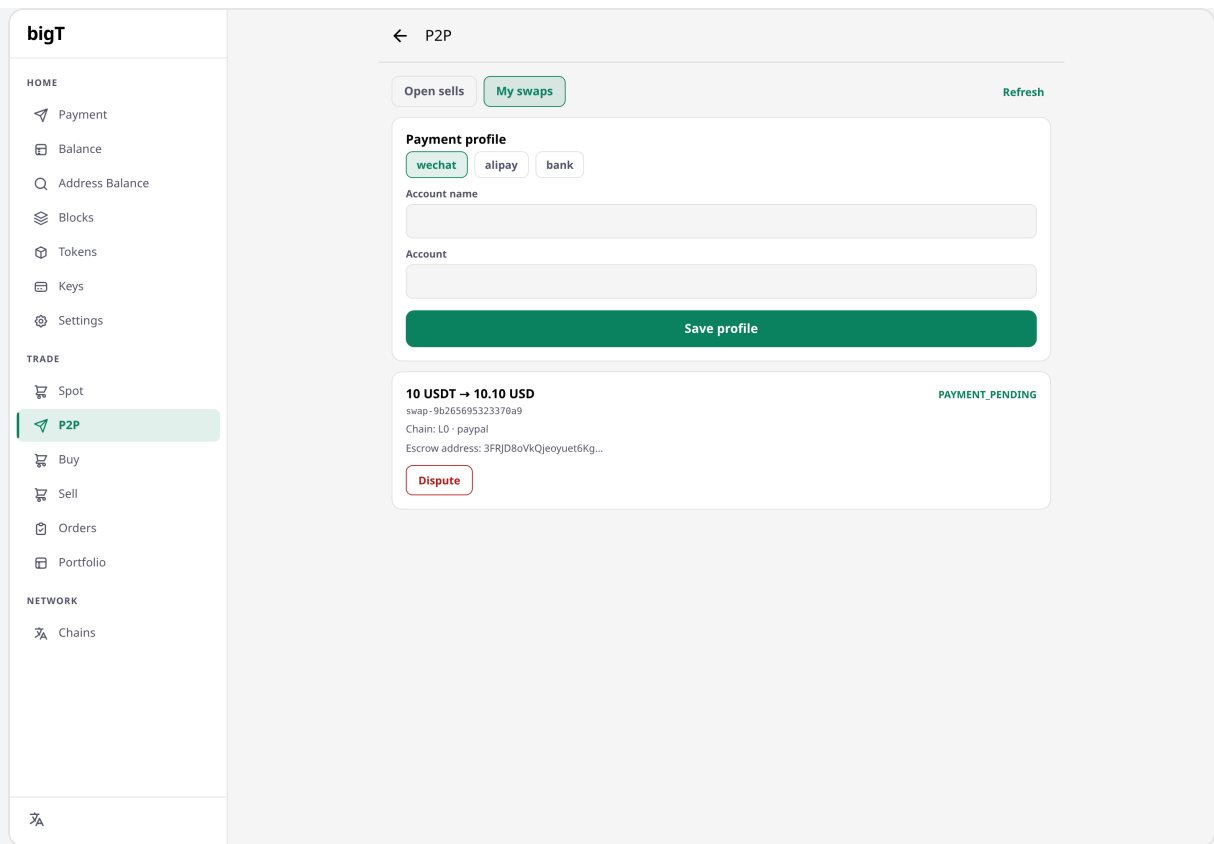
The buyer sees the funds secured **before** sending any fiat, and gets an

I have paid action once the lock is proven:



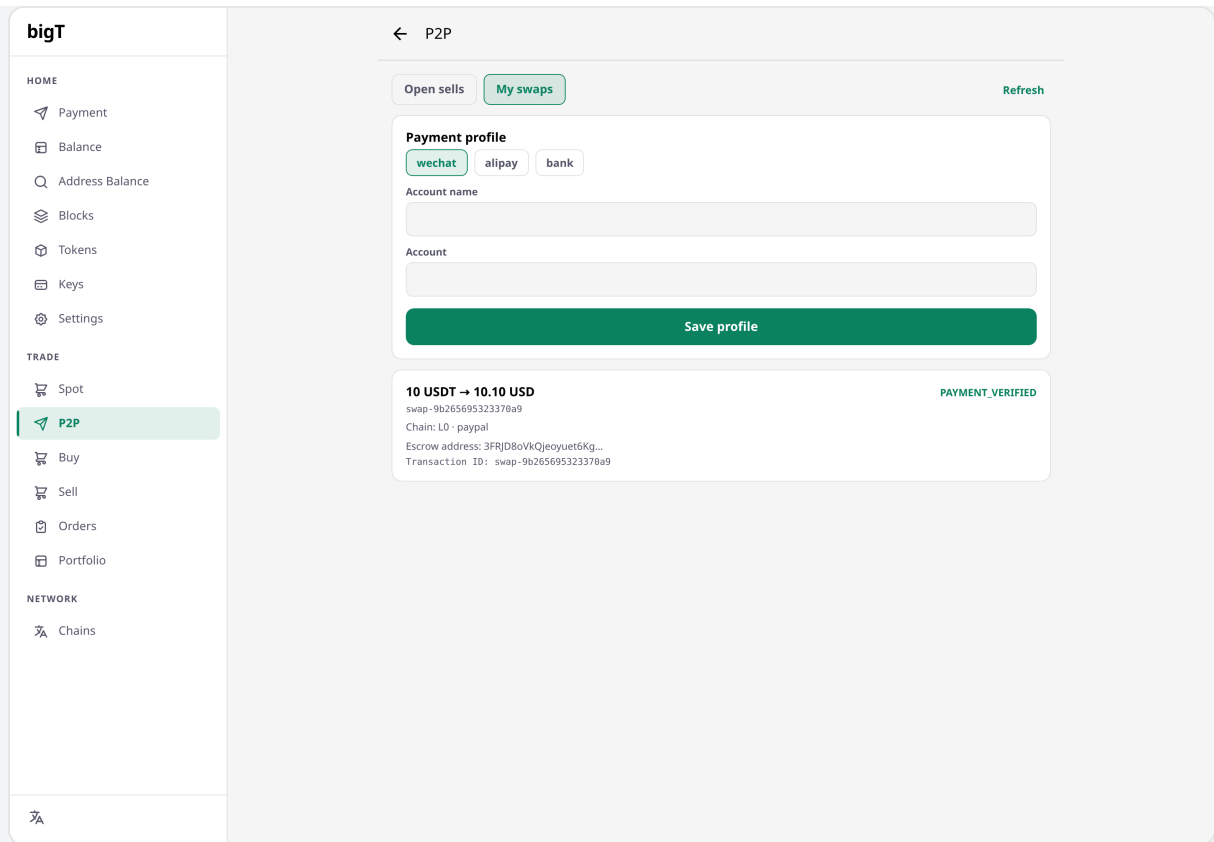
4. The buyer pays

The buyer pays the hosted PayPal invoice (PayPal holds the money – the engine never does) and reports the payment. This is only a *hint*; the real verification is the engine's own evidence. The swap is `PAYMENT_PENDING`, and a timeout timer starts – if the invoice is never paid, the seller can expire and refund the escrow without the buyer's consent:



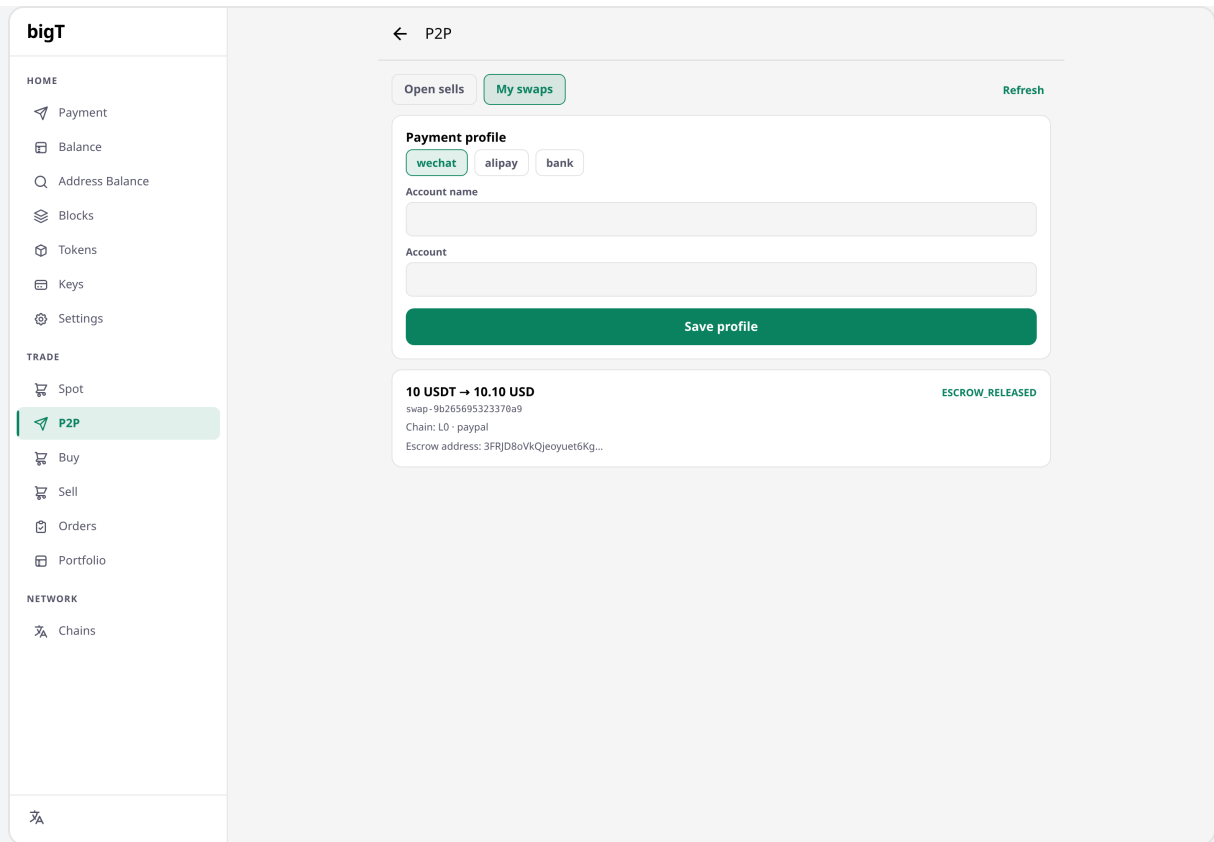
5. The engine verifies the payment

No seller confirmation is involved. The engine checks the payment from its own source and advances the swap to `PAYMENT_VERIFIED` – the escrow is now ready to release to the buyer's receive address:



6. Release – the funds move

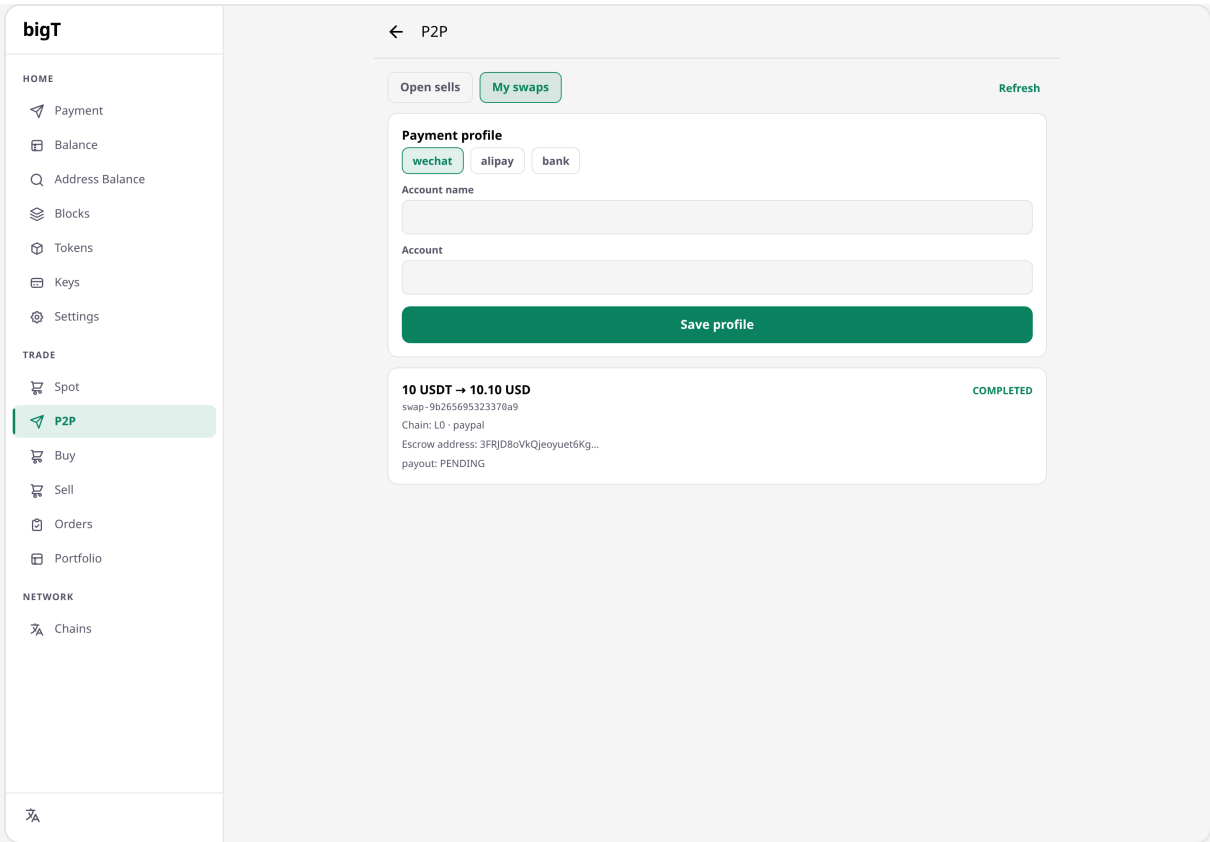
Release moves the escrowed tokens: the engine and the buyer each sign the spend from the escrow output, and two signatures satisfy the 2-of-3 script. The tokens land at the buyer's receive address and the swap is `ESCROW_RELEASED` :



7. Payout – the seller gets the fiat

The final step pays the seller through PayPal Payouts, and the swap reaches

COMPLETED :



The payout outcome (SUCCESS / FAILED / HELD / ONHOLD) arrives from

PayPal's webhook or is polled as a fallback; a failure can be retried from

COMPLETED without redoing the trade.

What protects you

Risk	Mitigation
Counterparty walks away	Funds sit in a 2-of-3 P2SH address; nobody can move them alone
Under- or over-payment	The invoice is exact-amount, so it either pays in full or stays unpaid

Fake payment claim	The engine verifies the payment itself – the payer cannot self-assert it
Engine goes rogue	Engine-only steps require the engine DID signature; every transition is anchored and publicly auditable
Chargeback after release	PAYMENT.CAPTURE.REVERSED sets a freeze: forward steps stop, refund/expire stay reachable
Dispute	CUSTOMER.DISPUTE.* pauses the swap until it resolves
Payout failure	HELD / FAILED / BLOCKED are first-class states; retry with the same payout reference

Going live

The demo runs against the in-repo settlement service with mock PayPal. To run the real flow you need a PayPal business account with Payouts activated, its API credentials and webhook id, and the escrow signer wired to the L0 broadcast step. Until then the engine runs the same state machine without touching real money.